

[Hi iam shri]

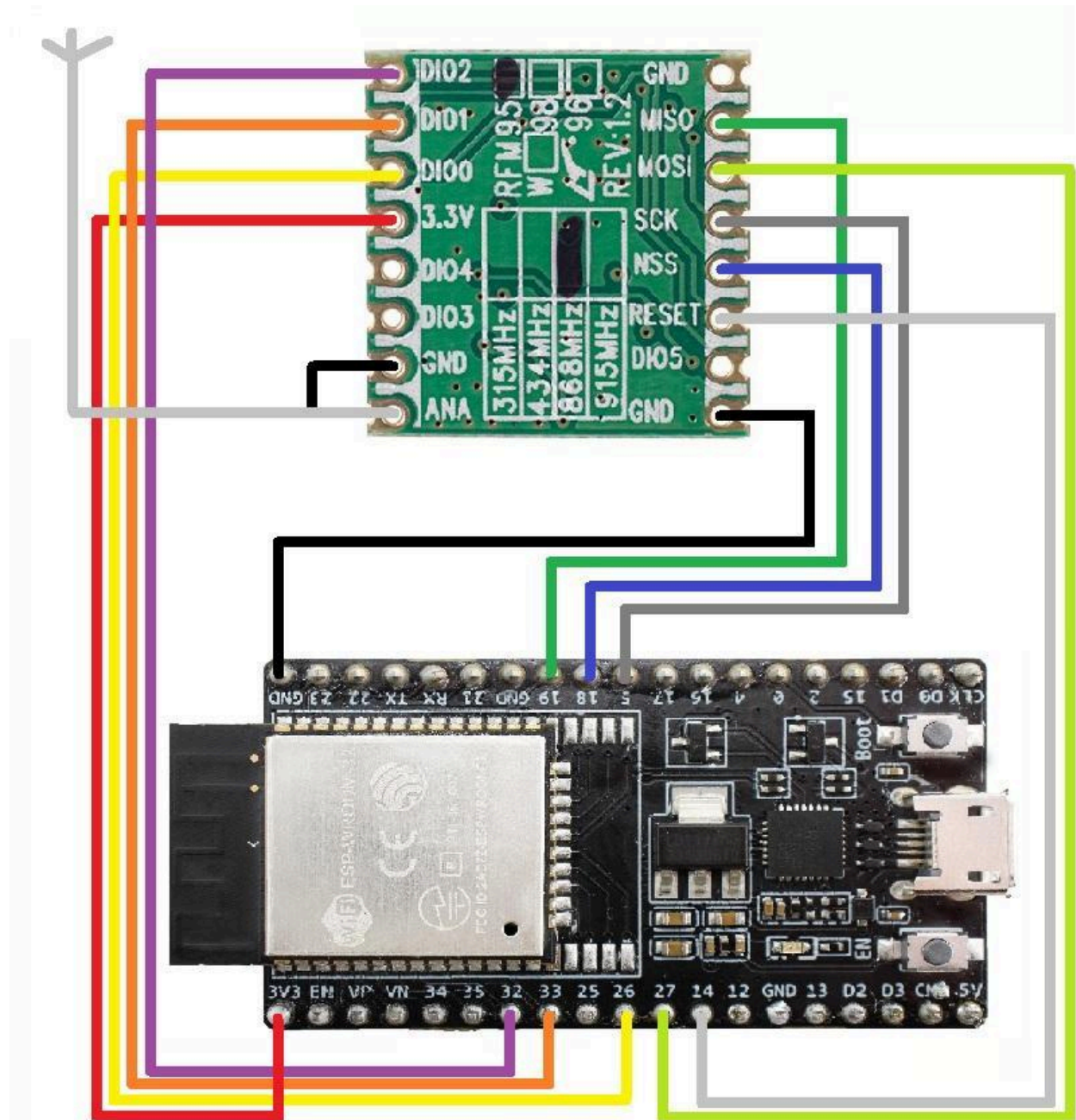
Using **ESP32 + SX126X LoRa HAT** is a very powerful setup for long-distance wireless communication.

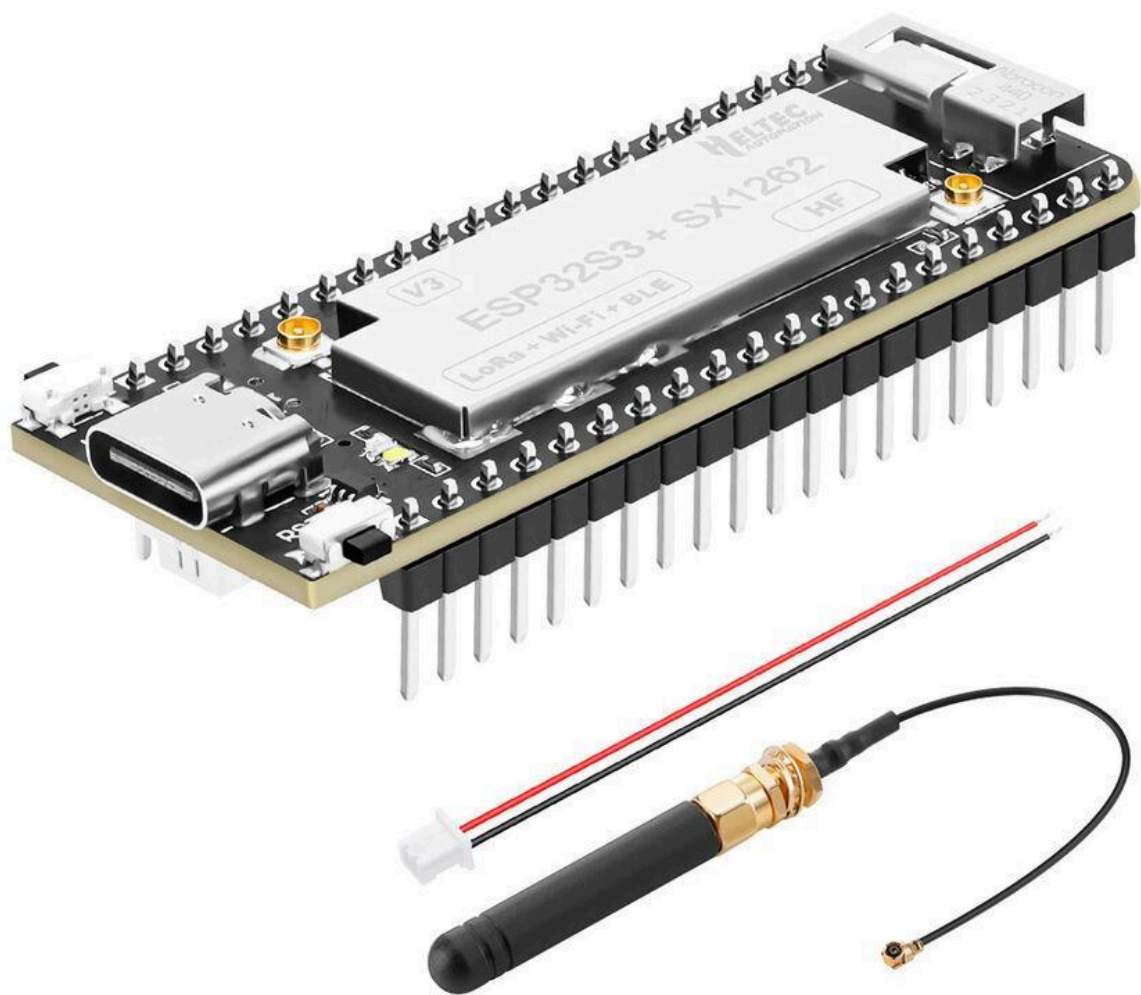
You can easily send:

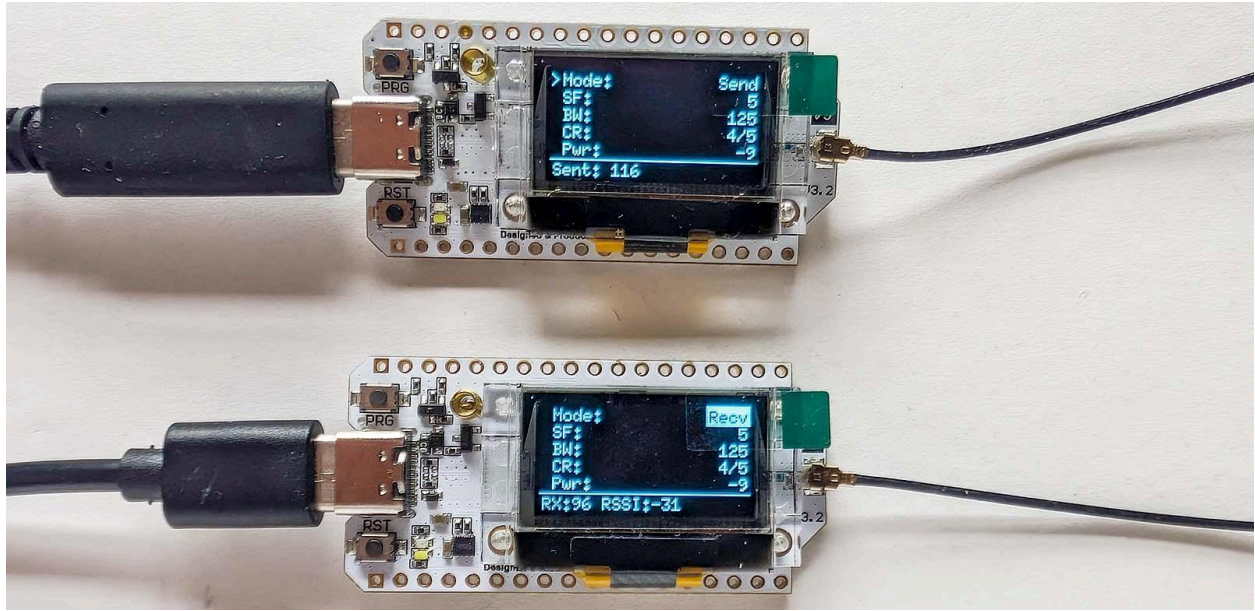
- text messages
- sensor data
- GPS location
- images (small packets)
- IoT data

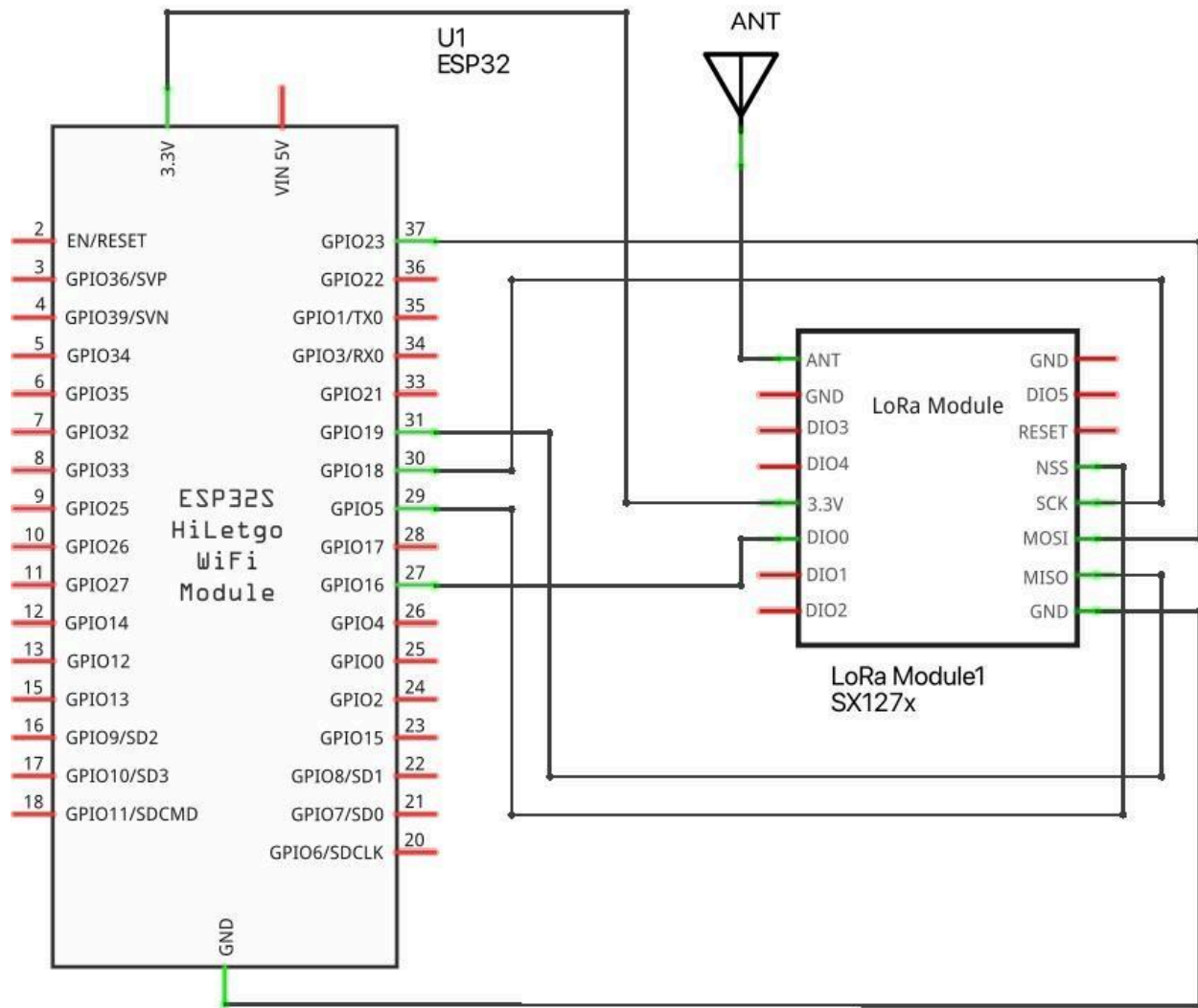
for **2 km to 10+ km** depending on antenna and environment.

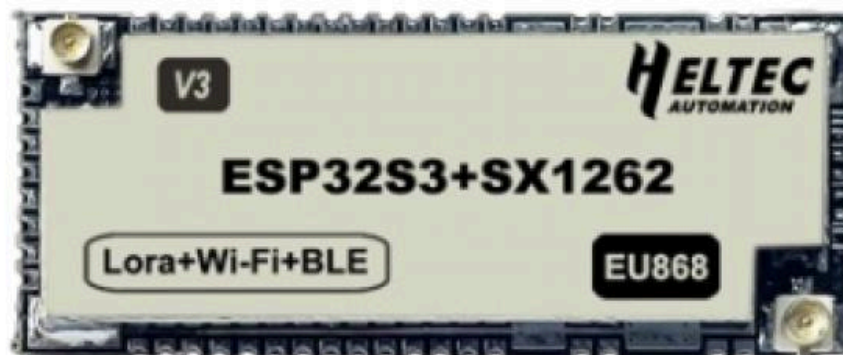
Components Needed

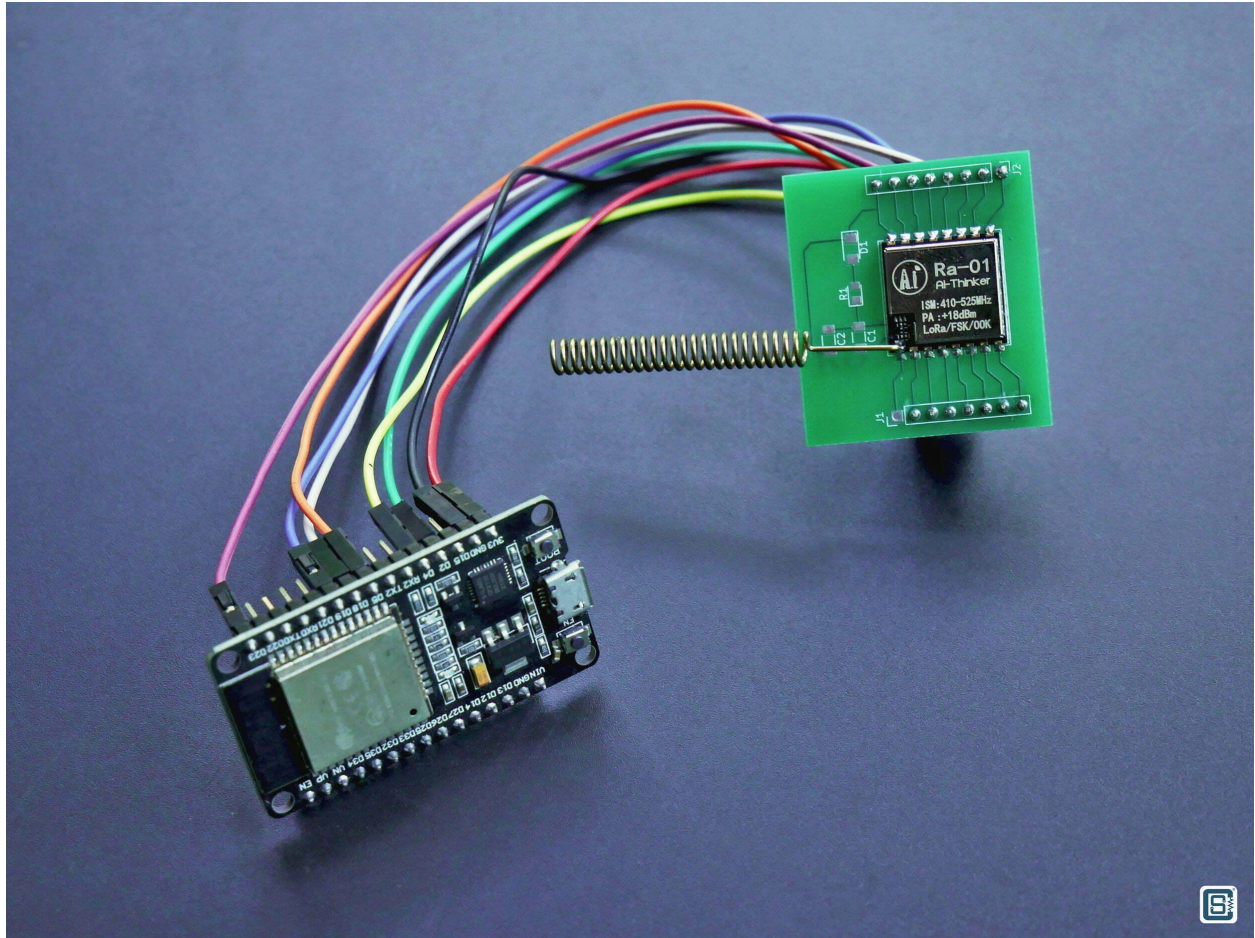












LoRa 32 V3 + 5dBi Antenna



- 2 × ESP32 boards
- 2 × SX1262 / SX1268 LoRa modules
- antennas
- jumper wires

Wiring (ESP32 ↔ SX1262)

SX1262 ESP32

VCC	3.3V
GND	GND
SCK	GPIO18
MISO	GPIO19
MOSI	GPIO23
NSS/C S	GPIO5
RST	GPIO14
BUSY	GPIO25
DIO1	GPIO26

 Important:

- Use **3.3V only**
 - Connect antenna before power ON
-

Install Arduino IDE Libraries

Open Arduino IDE.

Install:

RadioLib

Library author:

- Jan Gromes
-

Sender Code (ESP32)

```

#include <RadioLib.h>

// NSS, DIO1, RESET, BUSY
SX1262 radio = new Module(5, 26, 14, 25);

void setup() {

  Serial.begin(9600);

  Serial.print("Starting LoRa ... ");

  int state = radio.begin(865.0);

  if (state == RADIOLIB_ERR_NONE) {
    Serial.println("success!");
  } else {
    Serial.print("failed, code ");
    Serial.println(state);
    while (true);
  }
}

void loop() {

  String msg = "Hello from ESP32";

  int state = radio.transmit(msg);

  if (state == RADIOLIB_ERR_NONE) {
    Serial.println("Message sent!");
  } else {
    Serial.println("Send failed");
  }

  delay(2000);
}

```

Receiver Code (ESP32)

```

#include <RadioLib.h>

```

```

// NSS, DIO1, RESET, BUSY
SX1262 radio = new Module(5, 26, 14, 25);

void setup() {

  Serial.begin(9600);

  Serial.print("Starting LoRa ... ");

  int state = radio.begin(865.0);

  if (state == RADIOLIB_ERR_NONE) {
    Serial.println("success!");
  } else {
    Serial.print("failed, code ");
    Serial.println(state);
    while (true);
  }
}

void loop() {

  String str;

  int state = radio.receive(str);

  if (state == RADIOLIB_ERR_NONE) {

    Serial.print("Received: ");
    Serial.println(str);

  }
}

```

Upload Steps

1. Connect ESP32
2. Select board:
 - ESP32 Dev Module
3. Select COM port
4. Upload code

5. Open Serial Monitor
 6. Set baud rate:
 - 9600
-

Range Tips

For 2 KM+

Use Good Antenna

Very important.

Keep Antenna Vertical

Improves signal.

Open Area

Less buildings/trees.

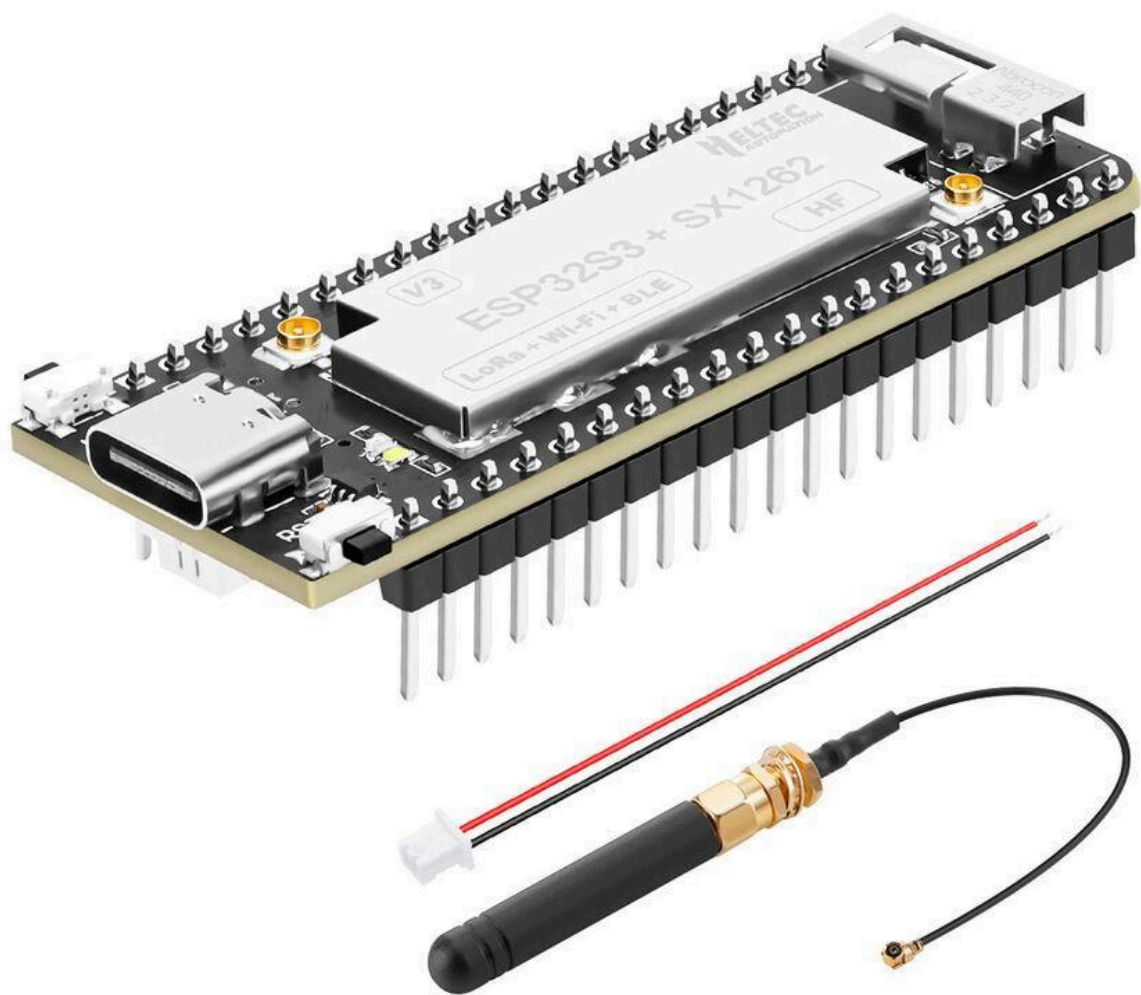
Use Low Bandwidth

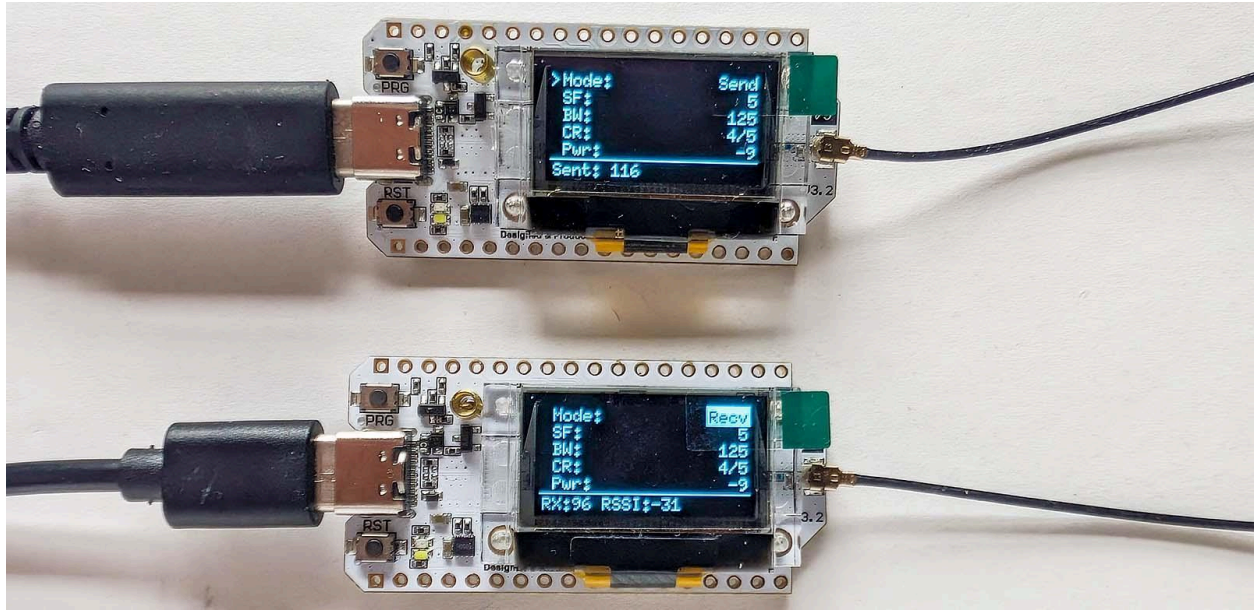
Longer range.

Expected Distance

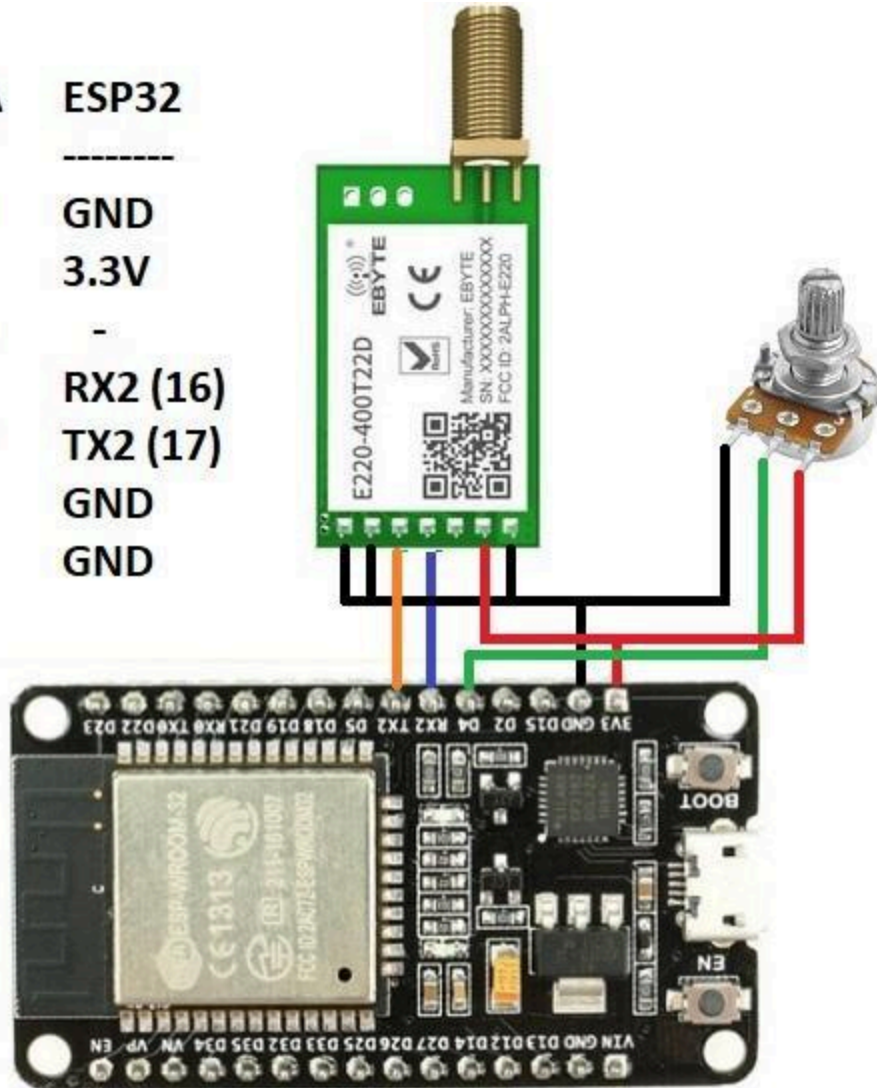
Area	Range
City	2–5 km
Open field	10+ km

Communication Diagram





LORA	ESP32
-----	-----
GND	GND
VCC	3.3V
AUX	-
TXD	RX2 (16)
RXD	TX2 (17)
M1	GND
M0	GND

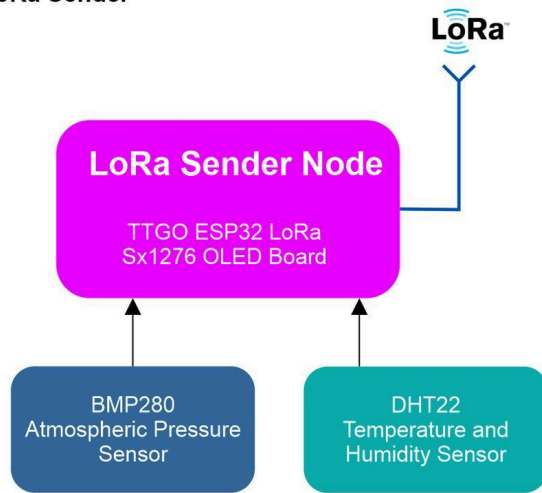




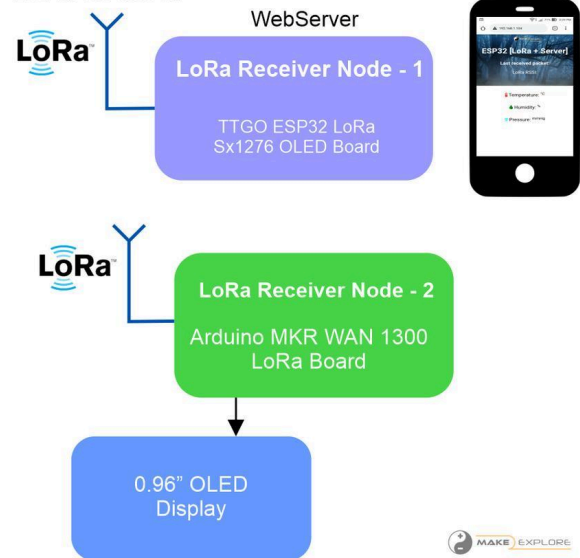
Block Diagram

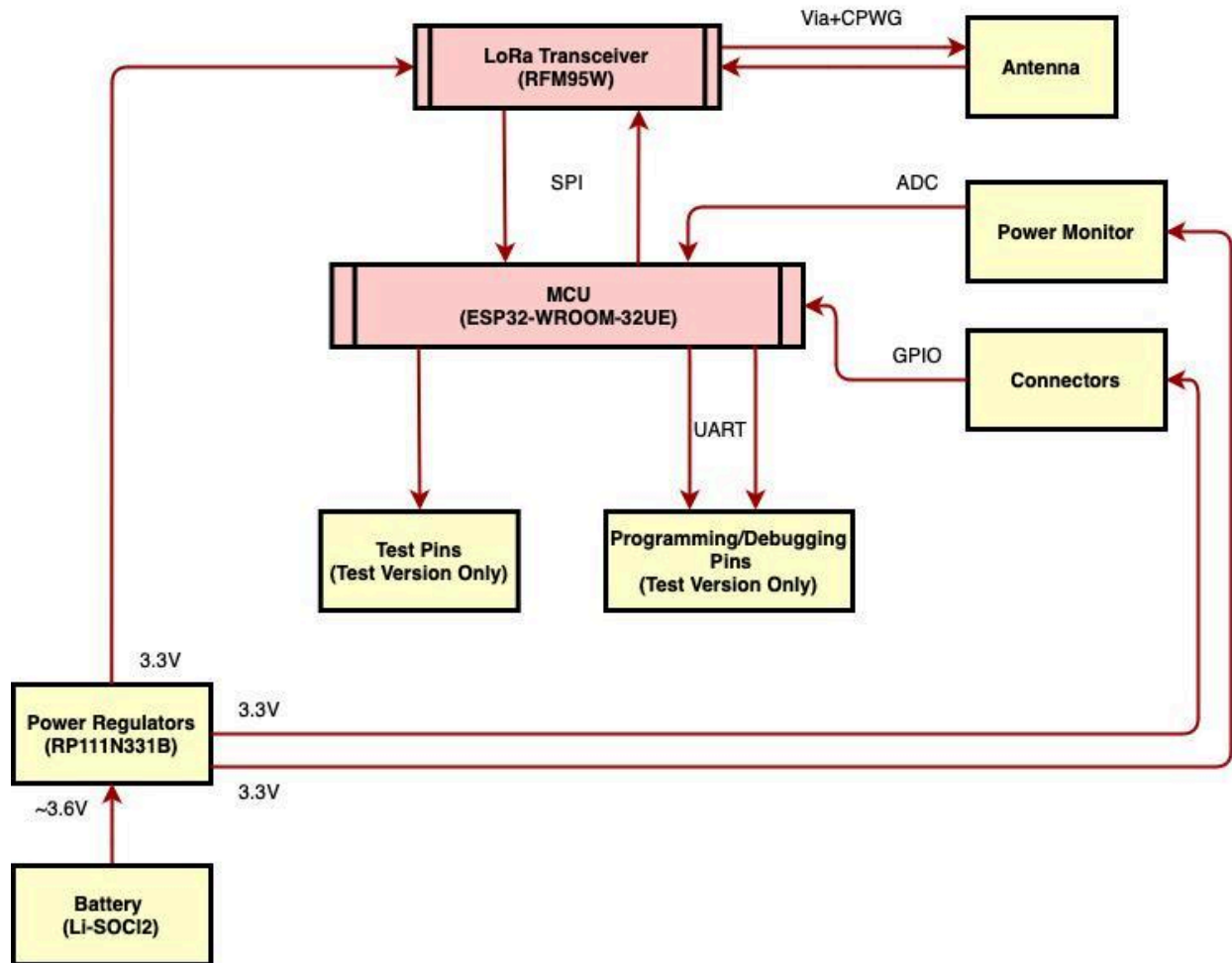
LoRa Remote Sensor Monitoring with web-server using
ESP32 LoRa and Arduino MKR WAN 1300

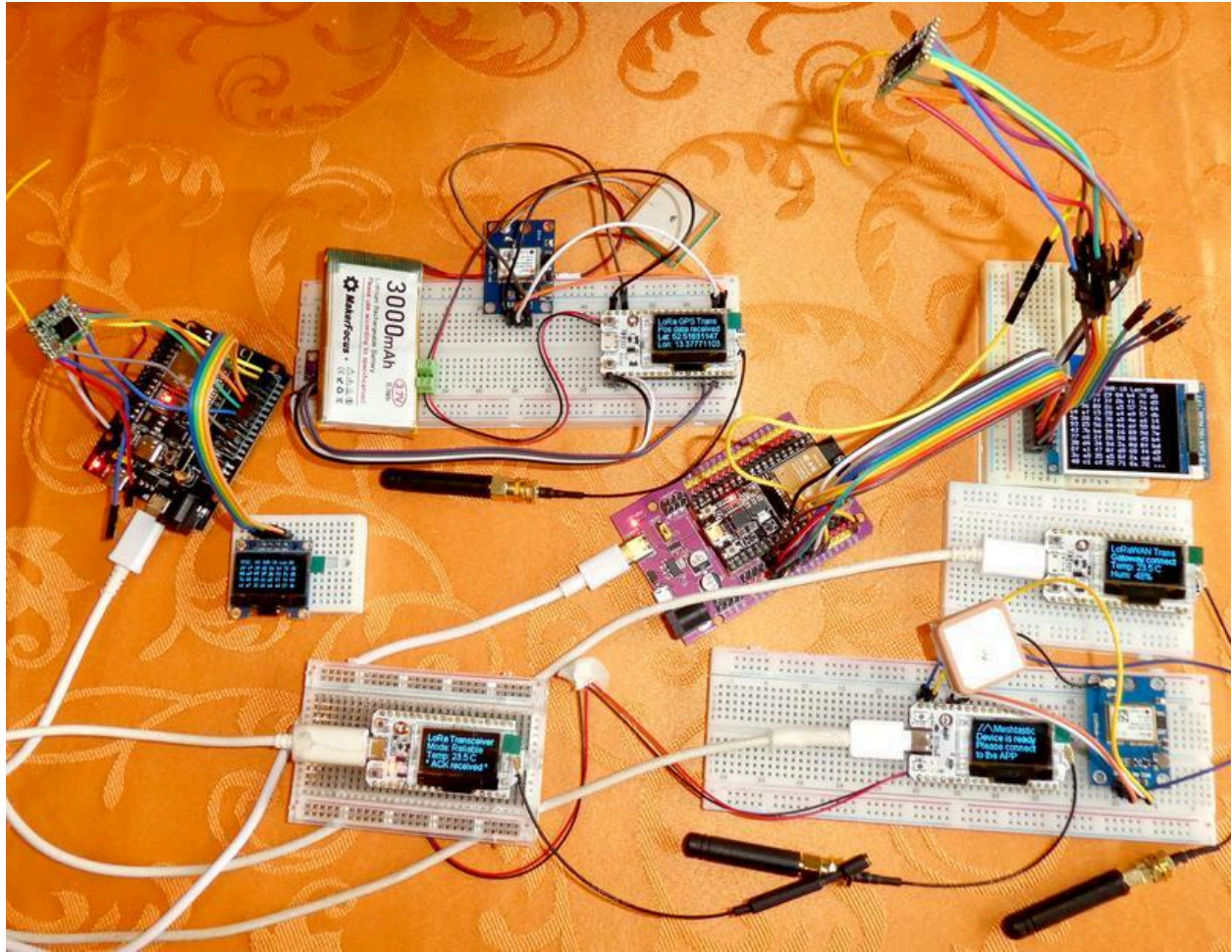
LoRa Sender



LoRa Receiver







ESP32 ← LoRa → ESP32
Sender Receiver

Advanced Projects

With ESP32 + SX1262 you can build:

- wireless CCTV alerts
- AI sensor nodes
- GPS tracking
- drone telemetry
- off-grid chat device
- disaster communication system
- Meshtastic node

Very powerful combination for Embedded + IoT + AI projects 🚀